



Investigation of the single-phase immersion cold plate amid PAO-4 and Noah@3000A – An experimental approach and its numerical verification

Yong-Dong Zhang, Yu-Chi Lin, Chi-Chuan Wang^{*}

Department of Mechanical Engineering, National Yang Ming Chiao Tung University, Hsinchu, Taiwan

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ABSTRACT

This study investigates improving traditional immersion cooling systems by employing cold plates in single-phase immersion cooling with dielectric fluids (Noah@3000 A and PAO-4). Experimental and CFD methods explore various cold plate structures in a 1 U server with a TTV heat source. Flow rates range from 1 to 3 LPM, and configurations include side-open, fully enclosed, and dual-open designs. Results show that incorporating cold plates reduces fluid bypass, enhancing thermal performance. High-viscosity dielectric oil as a coolant yields substantial improvements (36.1% and 41.6% at low and high flow rates) compared to traditional systems. Thermal resistance decreases with higher flow rates, more notably for Noah@3000A. Immersing the cold plate in liquid enhances convective heat transfer, reducing thermal resistances by 7.2% and 9.4% for Noah@3000A and PAO-4 at low flow rates. Smaller fin pitch consistently lowers resistance by eliminating bypass. Outlet configuration significantly influences thermal resistance, with Enclosed cold plates showing the least performance for PAO-4 and dual-open cold plate yields the worst performance with Noah@3000A. Differences in heat transfer behavior between PAO-4 and Noah@3000A are linked to laminar and turbulent flow characteristics. These findings offer insights into optimizing immersion cooling systems, emphasizing the impact of fluid choice and cold plate design on thermal performance.

1. Introduction

The rapid development of artificial intelligence (AI) big data, high performance computing (HPC) and other technologies raises huge demand and dramatic growth of datacenters. It is predicted that the energy consumption of data centers will become even more severe in the coming years [1–4]. Of the electrical energy consumption in the data center, 40% of the energy consumption is used for cooling [5]. The reliability of datacenter operation depends on cooling performance [6,7]. For datacenters, which are the backbone of the IT industry, require intensive cooling designs to facilitate their functionality. The cooling system acts not only to relieve thermal power but also to improve the IT system energy efficiency considerably [8]. The most common cooling method for datacenters is air cooling [9,10]. However, air cooling systems are an inefficient method of cooling as far as system efficiency is concerned. Huge air volumetric flowrate with tremendous fan power consumption accompanied with air cooling technology, thereby causing lower system efficiency and noise problems [11,12]. In this regard, liquid cooling offers significant advantages over traditional

air cooling, and is becoming more popularly recently [13,14]. The liquid cooling can be made available indirectly (cold plate) or directly (immersion cooling).

Among them, immersion cooling can handle high-flux electronic devices [15] for it immerses the electronics in a dielectric fluid [16] for heat removal. Immersion cooling contain superior heat transfer performance, low cost (if use dielectric oil), simple structure, high reliability and environmental friendliness, and is considered to be one of the most promising liquid cooling systems [17,18]. Qiu et al. [19] found that the cooling capacity of water immersion cooling system is 10 times that of forced air cooling, and even natural immersion cooling is much more effective than air cooling. Kanbur et al. [20] conducted thermodynamic and economic evaluations of single-phase and two-phase immersion cooling in datacenter systems with different power levels. The research found that the performance of the two-phase cooling system was 70% higher than that of the single-phase cooling system. However, the cost of a two-phase cooling system is nearly five times that of a single-phase cooling system. In addition, using indirect two-phase cooling via cold plate may lead to severe pressure drop oscillations and mild parallel

^{*} Corresponding author.

E-mail address: [ccwang@nycu.edu.tw](mailto:cwang@nycu.edu.tw) (C.-C. Wang).

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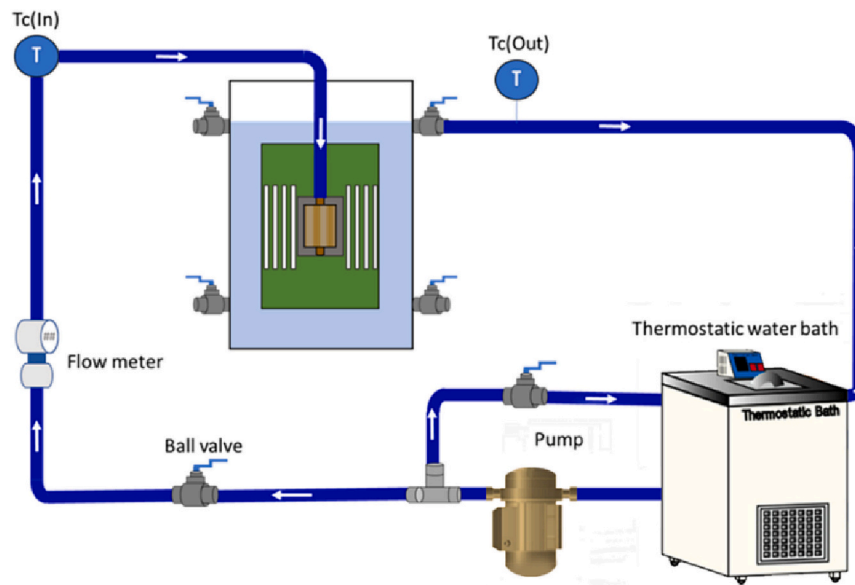


Fig. 1. Schematic diagram of the experimental rig.

channel instability due to hydrodynamic instability [21,22]. Hence, single-phase fluid immersion cooling becomes one of the best options for stability control, and high performance with affordable cost. Bansode et al. [23] verified that fully submerged servers can operate efficiently under extreme temperatures. There were numerous studies indicating that single-phase immersion cooling systems contain to improved power usage effectiveness (PUE). Compared with the natural circulation single-phase immersion cooling system, the forced convection cooling system has better performance [24,25]. Taddeo et al. [26] used experiments and numerical analysis to verify that single-phase immersion cooling is a more effective technique to cool server tank consisting of 54 servers with 400 W for each server. Due to electrical hazard considerations, advanced immersion cooling systems normally incorporates dielectric liquids as the working fluid [27].

At present, the most common dielectric liquids include fluorinated liquid and oil. Cheng et al. [16] used 3M Novec 7100 as coolant to study the cooling performance of a single-phase immersion cooling system under three circulation rates and two heat sinks having different materials. The results show that the heat dissipation effect depends on the circulating speed of the liquid coolant. If the circulation speed of the liquid coolant is low, a high temperature prevails along with highly uneven heat load distribution. Notably, the results indicate that the material of the heat sink imposes no significant effect on the results. Shrigondekar et al. [28] tested the single-phase immersion for fluorinated liquid (FC-40) and dielectric oil (PAO-6). The results show that the heat dissipation performance of FC-40 is better than that of dielectric oil (PAO-6) for its much lower viscosity. From the overall considerations including heat dissipation performance, cost and environmental benign, oil may be superior to fluorinated fluid immersion cooling [29,30]. In addition, because of its higher reliability, high heat dissipation and performance efficiency, single-phase oil immersed cooling technology may become viable for upcoming high-flux datacenter cooling [31,32]. In the oil-immersed cooling system, the change of kinematic viscosity of mineral oil is considered to be an important factor affecting the thermal performance of the cooling system [33]. Therefore, the subsequent work for immersed oil cooling should include optimizing heat sinks applicable for oil [34]. In order to improve the heat transfer coefficient of oil, Luo et al. [35] prepared oil-based nanofluids and analyzed the heat transfer performance for the Reynolds number ranging from 250 to 3000. The results show that the heat transfer performance of the heat sink with nanofluids has been significantly improved. In terms of heat sink

optimization, Li et al. [36] used the response surface method (RSM) to establish correlation subject to seven parameters, including maximum temperature, pressure loss, fin height, and fin spacing, yet they also utilized the regression correlations to optimize server performance well. The influence of location of inlet and outlet and fluid flow rate on the performance and efficiency of immersion cooling is also reported [37,38]. In particular, changes in the position of components on the server PCB board and any slot removal can have a significant impact on the cooling effect of other electronic devices [39]. The influence can be found in [28,40], where a severe bypass effect may occur for dielectric oil with a high viscosity.

The change of pressure drop will affect the flow distribution. Especially at the main heat source, the narrow fin pitch set to increase the heat transfer area will bring serious flow resistance, resulting in a decrease in flowrate, thereby the main liquid flow does not head into the heat sink which is so called bypass effect. In this regard, control of the coolant distribution in the server becomes a huge challenge. By changing the overall structure of the heat sink, the influence of the bypass effect can be relieved to improve the heat dissipation. In this study, the authors propose an alternative way to remove the effect of bypass. The cold plate is adopted in association with immersion cooling system. The design uses indirect cold plate to avoid conventional bypass problem. This can be made available via forcing dielectric fluid through the cold plate which is placed on top of the heat source. Subsequently, the fluid out of the cold plate submerges the whole PCB board to facilitate secondary cooling. This paper compares the performance of cold plate immersion system and traditional immersion system, and introduces several cold plate structures suitable for immersion system. The studies analyze the heat distribution and flow field of the immersed cold plate system in dielectric fluid (Noah@3000 A) and dielectric oil (PAO-4) through experimental investigation along with numerical verification. In addition, the performance differences are compared between dielectric fluids of Noah@3000 A and PAO-4. It is expected to provide a more efficient solution for cooling of high-flux data centers.

2. Experimental setup

2.1. Experimental system framework

As shown in Fig. 1, the temperature of the fluid entering the test section is controlled by a constant temperature bath (Brand: B403H,

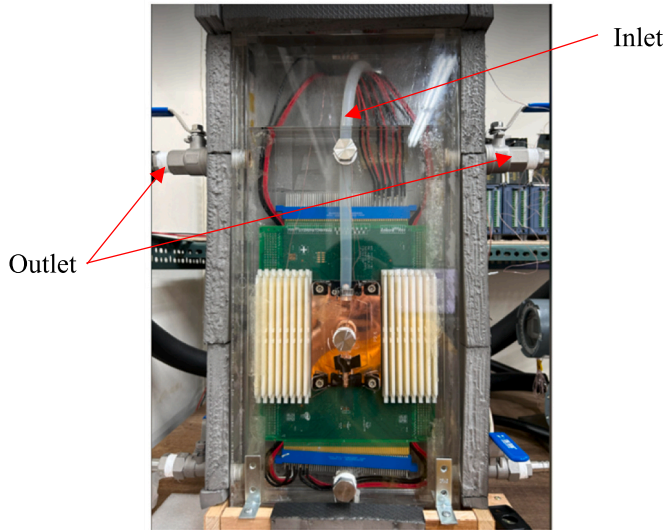


Fig. 2. Image of immersion cooling system.

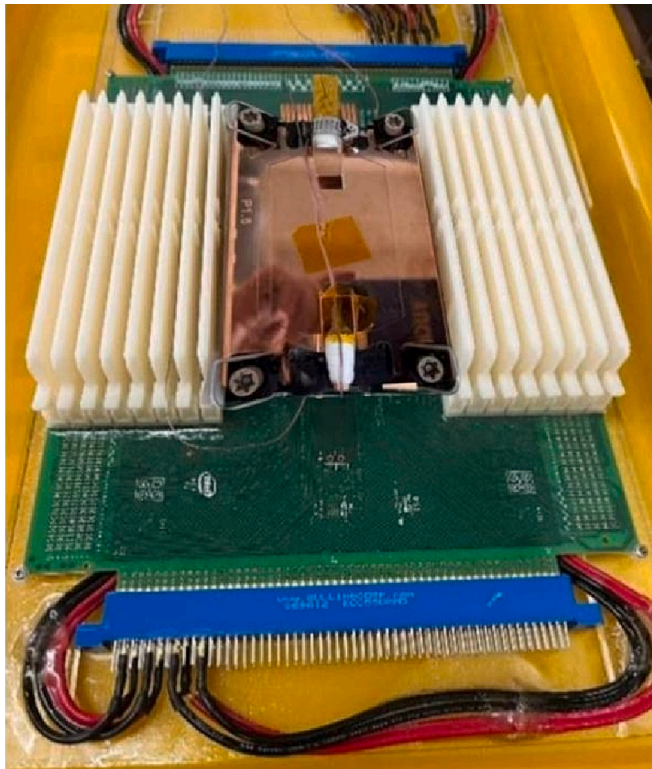


Fig. 3. TTV on the PCB.

cooling power: 1.5 kW), maintaining a stable temperature of 35 °C under different test flowrates. The working fluid circulates is driven by a pump. A flow meter is installed before the test section (Brand: BNC Industrial BH20-0F5-BSP, with a range of 0.33 to 21.67 LPM and an accuracy of 0.5%) to measure the inlet flowrate. The flow rate to the test section is controlled by a ball valve with bypassing pump liquid line.

The test tank (1U) is fabricated from a transparent acrylic (PMMA) material, allowing users to visually observe the internal flow behavior during experimentation. The enclosure's overall dimensions are designed to be 515 × 240 × 75 mm (length, width, and height), with a wall thickness of 15 mm. An internal tubing is used to direct the working fluid passing the test cold plate. The actual experimental setup is



Fig. 4. Case temperature of the TTV measuring point.

depicted in Fig. 2, which showcases a 1 U server immersed in the working fluid.

As illustrated in Fig. 3, a thermal test vehicle (TTV) is mounted on the PCB. Since the thermal load of the RAM is much lower than that of the CPU, a resin material crafted through 3D printing represents the RAM in this study. The CPU, situated beneath the cold plate, is substituted by a TTV to mimic an Intel CPU which is capable of generating equivalent thermal loads. The cold plate is installed atop the TTV, with the lower and upper pins of the PCB utilized to link the direct current (DC) source responsible for heating of the TTV. The total power is 300 W. The thermal resistance at the interface between the cold plate and TTV is reduced by applying a thermal interface material (thermal grease: TC-5888, thermal conductivity: 5.2 W/m-K, thickness ≤ 0.02 mm).

The input power ($Q = VI = 300W$) is assessed by recording the voltage (V) and current (I) from the DC power supply (6230 K-150), and T-type thermocouples are situated at the inlet and outlet of the cold plate. Another T-type thermocouple is located at the center of the TTV's upper surface, as indicated in Fig. 4, to gauge the cold plate temperature. The overall thermal resistance of the system is represented as follows:

$$R_{th} = \frac{T_{case} - T_{in}}{Q} \quad (1)$$

Among them, T_{in} is the inlet fluid temperature, T_{case} is case temperature of the TTV, Q is the total input power.

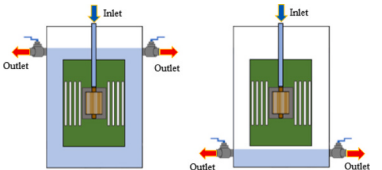
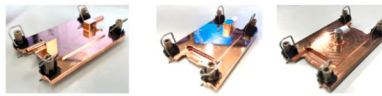
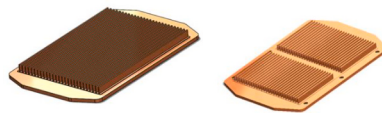
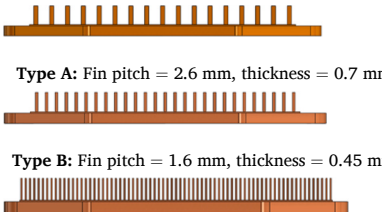
This study examines the influence of various parameters on system performance, including different outlets, fin pitch of the cold plate, flowrates, and the working fluids. Inside the cold plate, the Reynolds number across the fin channels ranges between 5 and 1100. The investigated system parameters in this study are listed in Table 1.

The measuring uncertainty of the T-type thermocouple, flowmeter, voltmeter, and ammeter is 0.1 °C, ±0.5%, ±0.05%, and ± 0.1%, respectively. The maximum uncertainty of evaluating system thermal resistance ranges from 1.43 to 2.47%. In evaluating the uncertainty, the precision error and the bias error of the proposed confidence of the experimental measurement are taken into account [41].

2.2. Working fluids

In this study, two distinct working fluids were examined. The first is Noah@3000 A, a dielectric fluid manufactured by Zhejiang Noah Fluoride Chemical Company in China, and the second one is the commonly used synthetic base oil, PAO-4. The variation in viscosity

Table 1
System parameters studied in the present work.

Parameter	Values/description and Illustration	Remarks
Immersion configuration		Compare the effect of cold plate with/without immersion in liquid on heat transfer.
Inlet/outlet configuration	Immersion  Enclosed cold plate Side-open cold plate Dual-open cold plate	The influence of inlet and outlet layout on heat transfer was studied. The central inlet has an impact cooling effect on the temperature measuring point (high temperature area). Dimensions: 117.96 mm × 77.96 mm × 23.83 mm (L × B × H)
Fin structure	 Integral type Segmented type $L_{total} = 60\text{ mm}, H = 3\text{ mm}$ $L_{total} = 54\text{ mm}, \text{Gap} = 6\text{ mm}, H = 3\text{ mm}$	Integral fins are used in both the Enclosed cold plate and the Side-open cold plate configurations. Impingement with split segmented fins are used for Dual-open cold plate.
Fin pitch of the cold plate	 Type A: Fin pitch = 2.6 mm, thickness = 0.7 mm Type B: Fin pitch = 1.6 mm, thickness = 0.45 mm Type C: Fin pitch = 0.6 mm, thickness = 0.2 mm	The fin spacing affects the heat transfer coefficient and pressure loss of the cold plate. The segmented fin is used for the impingement inlet is located at the middle of cold plate.
Flowrate	1, 2, and 3 LPM(Liters Per Minute)	One of the main parameters affecting heat transfer.

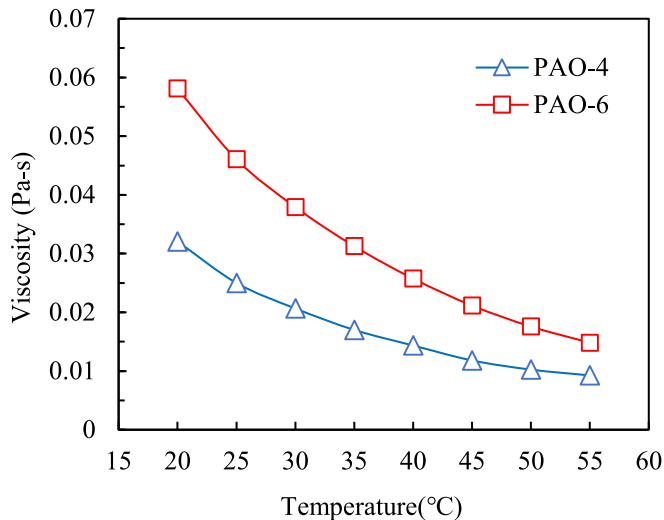


Fig. 5. Variation in viscosity of PAO-4 and PAO-6 with temperature.

with temperature for these fluids, compared to the frequently used dielectric fluid FC-40 and dielectric oil PAO-6, is depicted in Fig. 5 and Fig. 6, with additional parameters referenced in Table 2.

Fig. 5 and Fig. 6 demonstrate that the viscosity of all four working fluids significantly decreases as the temperature rises. In comparison

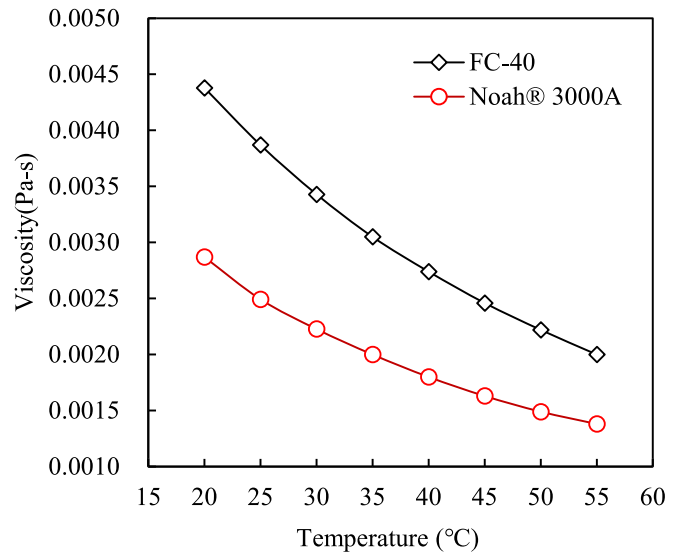


Fig. 6. Variation in viscosity of FC-40 and Noah@3000 A with temperature.

with FC-40, Noah@3000 A showcases a lower viscosity and better superior fluidity, thereby potentially enhancing its cooling performance. Notably, Noah@3000 A, which is also a fluorinated fluid, has an ODP (Ozone Depletion Potential) of zero and a GWP (Global Warming

Table 2
Major physical parameters of working fluid (At 35 °C).

Working fluid	Density (kg/m ³)	Thermal conductivity (W/m·K)	Specific heat (J/kg·K)	GWP
Noah® 3000A	1787	0.0614	1159	120
PAO-4	≈798	≈0.15	≈2143	0
FC-40 [40]	1828	0.0646	1068	9000
PAO-6 [40]	814	0.1452	2377	0

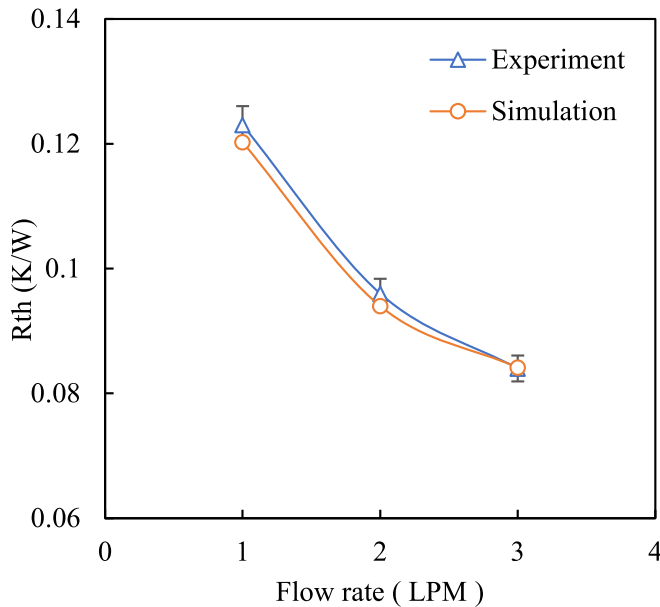


Fig. 7. Comparison between simulation results and experimental results. (Side-open cold plate: fin pitch = 1.6 mm, thickness = 0.45 mm, height = 3 mm, T_{in} = 35 °C, PAO-4).

Potential) of merely 120, as illustrated in Table 2, hence exerting a minimal impact on global warming.

Both PAO-4 and PAO-6 are synthetic base oils (poly-alpha olefin synthetic base oil), characterized by their low volatility and high thermal stability. PAO-4 is environmentally friendly with a GWP of zero and is substantially more cost-effective than fluorinated fluids. PAO-4 serves as a primary base oil for passenger car engines, heavy-duty diesel engines, transmission systems, gearboxes, and various industrial synthetic lubricants. Within the operational temperature range (35–55 °C), PAO-4

exhibits a lower viscosity than PAO-6, indicating better fluidity. The flash point of PAO-4 is 219 °C, slightly below that of PAO-6, which is 246 °C.

Table 2 lists Major physical parameters of the working fluid at 35 °C. As shown in the table, significant differences in properties such as density, thermal conductivity, and specific heat are observed between dielectric fluids and oils. At equivalent temperatures, the density of the dielectric oil is 2.1–2.3 times lower than that of the dielectric fluid, while the thermal conductivity and specific heat are more than two times higher. Compared to viscosity, the variations in density, thermal conductivity, and specific heat for the four working fluids within the operational temperature range are minimal.

2.3. Numerical simulation

In this investigation, Simcenter Flotherm XT is employed for numerical simulation to elucidate the experimental results. Because the flow field is complex, the turbulence model has been used to accommodate both laminar and turbulent flows, enabling the simulation to compute either type of fluid flow. The software employs the Navier–Stokes Equations to model both laminar and turbulent fluid flows and utilizes the k-ε model to address the transport equations governing turbulent kinetic energy and its dissipation rate [42,43]. Automated mesh sizing is applied, and localized refinement is performed at the fluid-solid interface. A mesh-independence test reveals that an optimal balance between accuracy and computational efficiency is achieved with approximately 3 million grid cells, which accurately depict system temperature and flow field characteristics.

As illustrated in Fig. 7, simulations employing this mesh configuration are compared with experimental results, demonstrating an excellent agreement with deviations between the simulations and experimental results within 2.3%.

3. Results and discussion

3.1. By-pass effect analysis of traditional cooling system

Fig. 8 is a schematic diagram of a traditional immersion cooling system. The system can be divided into 3 parts: CPU in the middle and RAM on both sides. The flow channel can also be divided into three parallel branches: branch-1, branch-2, and branch-3, respectively. Any structural changes may affect the flow distribution in each branch, which is called the bypass effect. Note that Fig. 9(a) is the schematic of flow distribution of the original structure, and the fin pitch of the heat sink is 1.6 mm. In order to improve the heat dissipation performance at the main heat source, the most common method is to reduce the fin pitch

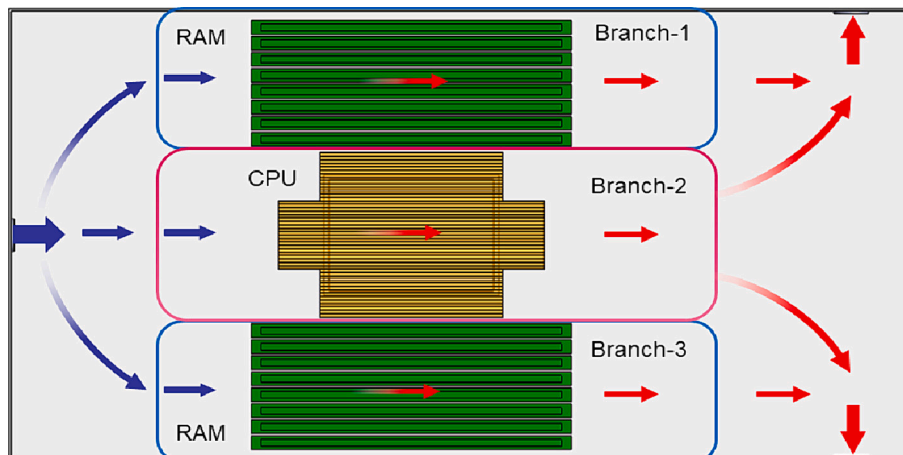
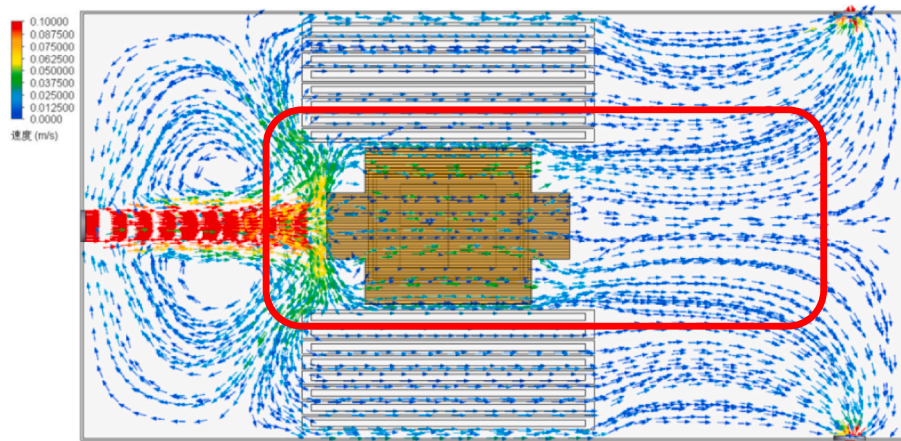
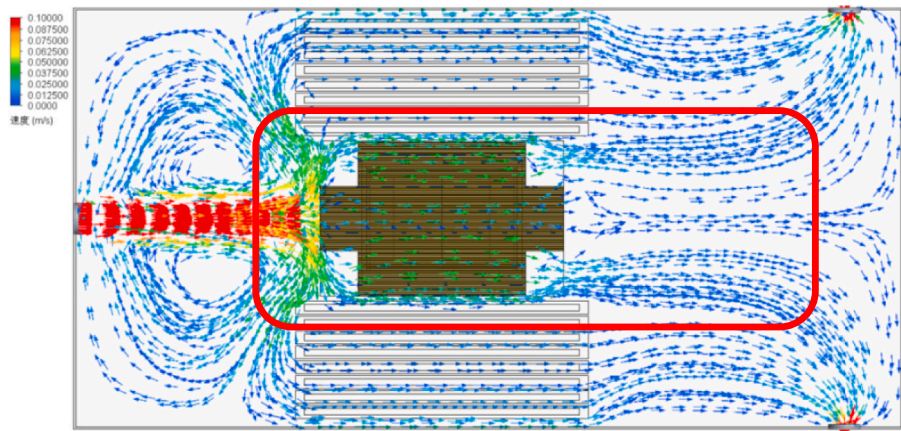


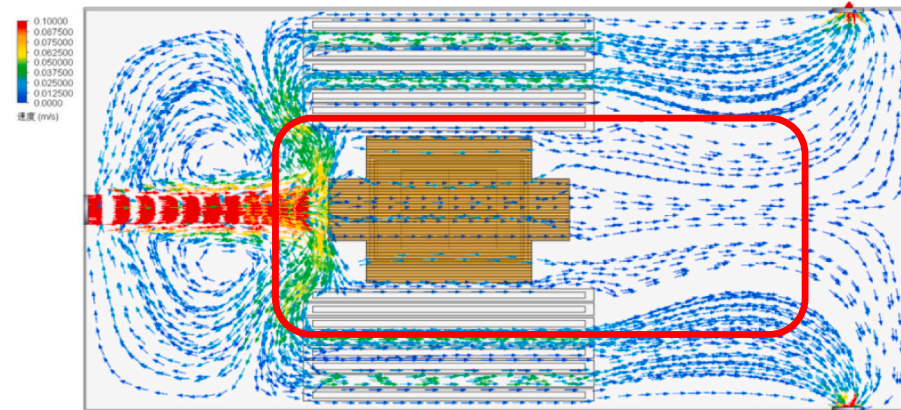
Fig. 8. Schematic diagram of the flow path of the traditional immersion cooling system.



(a) Original structure (Fin pitch = 1.6 mm)



(b) Densely fins (Fin pitch = 0.8 mm)



(c) Removed part of the RAM

Fig. 9. Effect of server structure change on fluid flow distribution.

to accommodate more surface area. However, the denser fins not only increase the heat dissipation area, but also increase the pressure drop of this branch, resulting in a decrease in severe bypass flowrate as depicted in Fig. 9(b). In this regard, the thermal performance may be jeopardized especially for a high viscous fluid like oil. Moreover, as shown in Fig. 9 (c), removing part of the RAM on both sides will also reduce the flowrate in branch-2.

In summary, the branches in the server are parallel, resulting in the bypass effect of traditional immersion cooling systems. In order to

reduce or even eliminate the bypass effect, this article uses a cold plate to replace the traditional air-cooling fins. This turns the parallel branch into a mixed branch structure with series connection first and followed by parallelization. In other words, the cooling fluid first flows into the cold plate and then submerges the system, as shown in Fig. 10. This design ensures that the flowrate through the main heat source will not be affected by changing other structures.

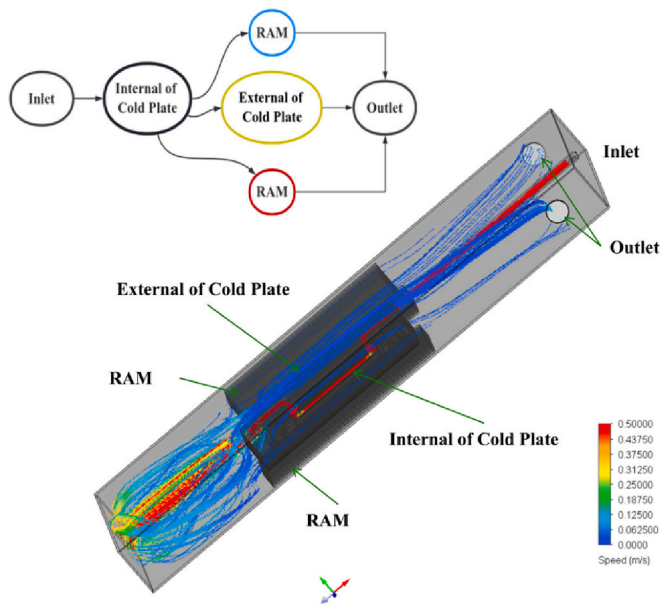


Fig. 10. Flow distribution of immersion cooling system with applied cold plate.

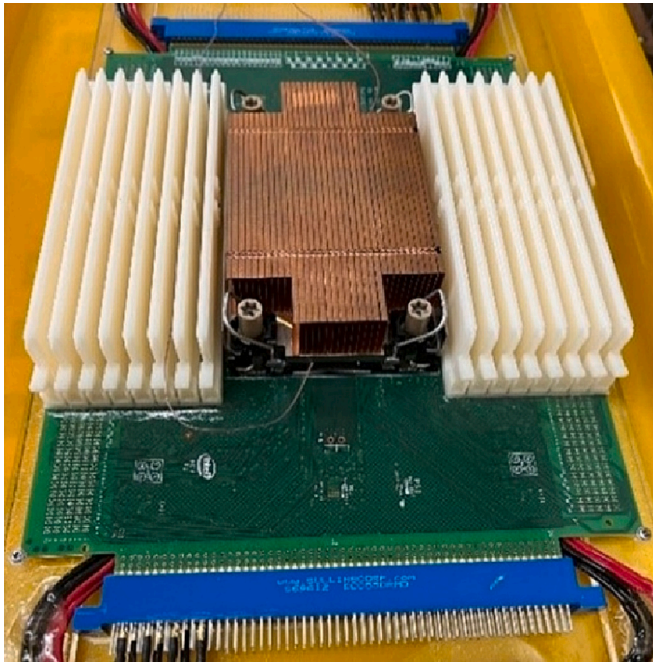


Fig. 11. Traditional air-cooled stack fins [28].

3.2. Comparison of liquid cold plate immersion and conventional immersion cooling systems

Traditional heat sink uses traditional air-cooled stack fins (Fig. 11). With an inlet temperature of 35 °C and the flow rate from 1 LPM to 3 LPM, the experiment compares the heat transfer performance between Enclosed cold plate (ECP) and traditional immersion cooling systems, as shown in Fig. 12. The working fluid is PAO-4. The dimensions of the fins for both systems are presented in Table 3.

As shown in the results, thermal resistance decreases with increasing flowrate for both systems under different working fluids. It is evident that the cold plate exhibits a more pronounced decline in thermal resistance with the rise of flowrate for Noah@3000 A. Specifically, when increasing from 1 LPM to 3 LPM, thermal resistance decreased by 39.7%

using Noah@3000 A and 31.4% using PAO-4 with the cold plate system. In contrast, the thermal resistance of the traditional heat sink exhibited marginal changes under Noah@3000 A and FC-40 dielectric fluids with increasing flow rates. The reason for the comparatively large drop is associated with the cold plate structure. Note that the flow into the cold plate with a sharp expansion with an effective Reynolds number of 4690 and 269 for Noah@3000 A and PAO-4 at 1 LPM, respectively. This indicates that the inlet flow condition for Noah@3000 is not laminar before hitting the fin surface. In this sense, the drop of thermal resistance is comparatively large when raising the flowrate. This can be made clear from the fundamentals of flat plate or channel flow where $Nu \sim Re^{0.8}$ for turbulent flow and $Nu \sim Re^{0.5}$ for laminar flow. Consequently, one can see that the drop of thermal resistance for Noah@3000 A is more pronounced than that of PAO-4. It is interesting to know that the thermal resistance for PAO-4 at a low flowrate of 1 LPM is lower than that of Noah@3000 despite it contains a lower Nusselt number (10.86 and 7.32 for Noah@3000 A and PAO-4, respectively). However, its thermal conductivity is 250% higher than that of Noah@3000 A, thereby yielding a higher heat transfer coefficient for PAO-4. In contrast, the thermal resistance of Noah@3000 A is slightly lower than that of PAO-4 when the flowrate is raised to 3 LPM. This is again associated with the comparatively turbulent nature for Noah@3000 A and the turbulent intensity is augmented when raising the flowrate. In the meantime, the flow field of PAO-4 is far within the laminar flow regime even the flowrate is raised to 3 LPM ($Re = 665$). Additionally, it was found that when using Noah@3000 A dielectric fluid, the traditional air-cooled heat sink showed the lowest thermal resistance when compared to a cold plate. At a flow rate of 3 LPM, the thermal resistance of Noah@3000 A is reduced by 49% compared to that of PAO-6.

Even when switching to the higher-viscosity dielectric fluid FC-40, the traditional heat sink still outperforms the cold plate design. This is actually not surprised because the surface area of the traditional air-cooled heat sink is 19 times higher than that of the cold plate. Even though the traditional heat sink suffers appreciably with severe flow bypass, its surplus surface manages to compensate and results in a lower thermal resistance. The significant bypass suggests that the flow field inside the heat sink is laminar, therefore the slope of thermal resistance vs. flowrate is relatively linear as depicted in the figure. Nonetheless, for systems using dielectric oils, cold plate exhibited comparatively lower thermal resistance. With the viscosity difference exists between PAO-6 and PAO-4, the improvement in performance is still substantial for immersion cold plate for PAO-4 when compared to PAO-6 for conventional heat sink. This discrepancy is attributed to the high viscosity of PAO-6 which provides some pronounced fluid bypass effects.

3.3. Performance comparison between immersion and non-immersion cold plate

Utilizing liquid cold plates as a substitute for traditional heat sink effectively mitigates the bypass effect in cooling systems. For cold plate operated without liquid immersion cooling, the exterior surface temperature measurement of the cold plate is considerably higher than that of the surrounding fluid as shown in Fig. 13(a). This observation is confirmed by simulation results as presented in Fig. 13(b), suggesting that immersing the cold plate contains a comparatively low temperature. This is because immersed fluid offers a higher convection heat transfer and enhances the overall thermal performance accordingly. Specifically, the T_{case} of the cold plate is reduced from 70.2 °C (in air ambience) to 63.2 °C at the same inlet conditions.

To investigate the heat transfer effects when the cold plate is free from immersion experimentally, the exit of cold plate was connected with tubing toward the bottom of the PCB board, as shown in Table 1, which ensures that the fluid level remained below the cold plate while exposing its exterior to ambient air. Fig. 14 and Fig. 15 shows the comparisons of the thermal performance for cold plate with and without immersion for flowrate = 1 LPM and 3 LPM, respectively. It is shown

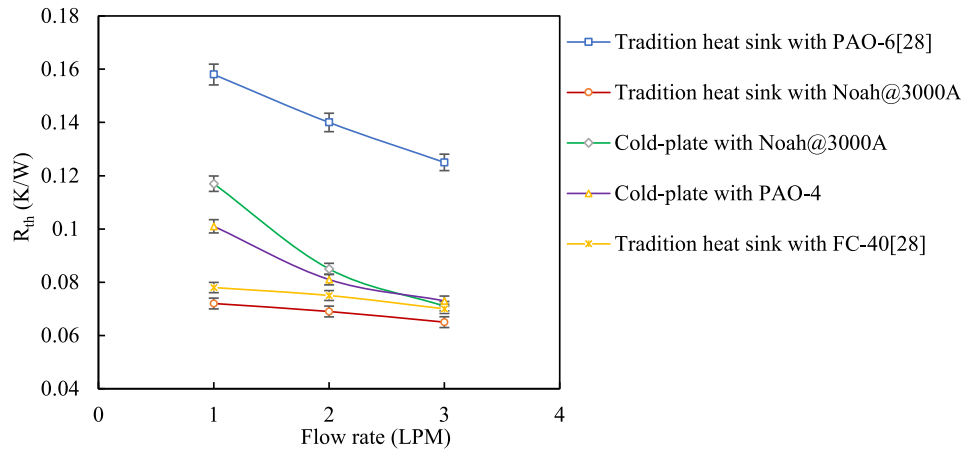


Fig. 12. Comparison of heat transfer performance of Traditional fins and cold plates using different working fluids.

Table 3

Structural parameters of the heat sink.

Type of heat sink	Fin pitch (mm)	Fin thickness (mm)	Fin height (mm)	Heat dissipation area (cm ²)
Traditional heat sink	1.2	0.25	22	26.6
Cold plate	1.6	0.45	3	1.4

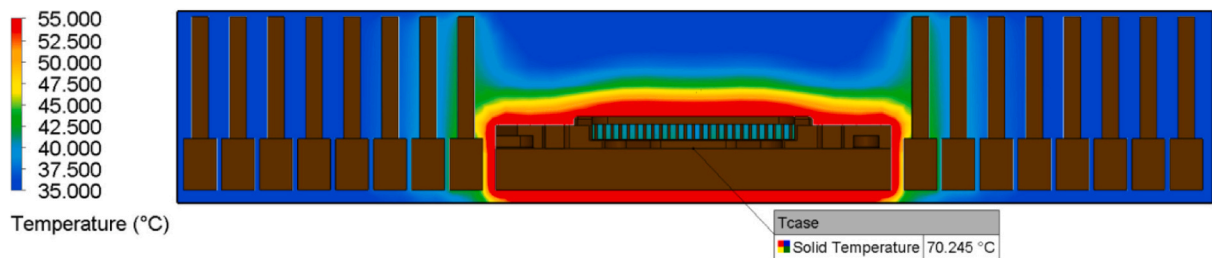
that under equivalent thermal power and inlet temperatures, the thermal resistance is notably lower when the cold plate that is immersed in liquid. These test results indicate that due to different heat transfer coefficients between the liquid and air, immersion of the cold plate fortifies external convection of the casing of cold plate, thus augmenting system performance. Specifically, at a flow rate of 1 LPM using Noah@3000 A as the working fluid, the system's thermal resistance decreased by 7.2% and by 9.4% when using PAO-4. At a flow rate of 3 LPM, the thermal resistance is decreased by 2.7% and 3.9%, respectively, for both fluids.

3.4. Performance of variations in cold plate design

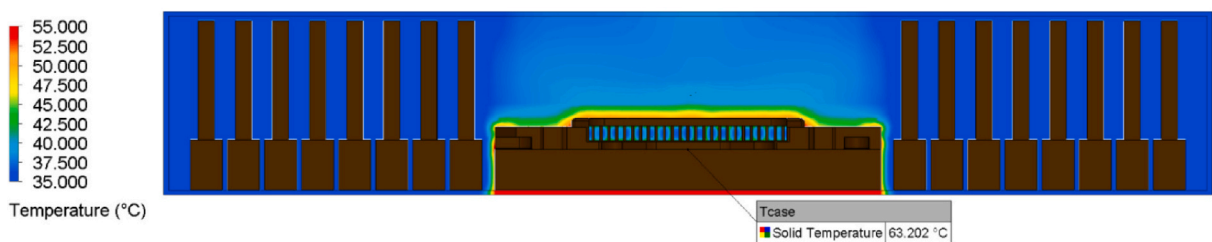
To assess the influence of the outlet configuration of the immersion

cold plate on system thermal performance, tests were conducted on three distinct cold plate designs: namely fully enclosed, side-open, and dual-open. Note that the inlets into the cold plate are with an impingement configuration. The internal fin spacing for all three designs was 1.6 mm, and the fin thickness was 0.45 mm. However, due to the inlet location is at the center of the Dual-open cold plate, segmented fins were utilized. These fins were symmetrically arranged around the central inlet, leaving a gap between them, thus shortening the overall fin length. The shapes of the cold plates and fins are detailed in Table 1. More details on the geometry of the impingement cold plate are shown in Fig. 16.

Experimental results of the three cold plates with Noah@3000 A and PAO-4 are depicted in Fig. 17 and Fig. 18, respectively. Regardless of the working fluid, the thermal resistance consistently decreased with increasing flowrates. However, the decline trend for Noah@3000 A is obviously more pronounced than that of PAO-4. This is again associated with the fluid flow characteristics of Noah@3000 A. As depicted earlier, the inlet Reynolds number Ranges from approximately 5000 to 12,000 for Noah@3000 A, meaning that the flow inside the cold plate behaves more like a turbulent flow nature. Hence, HTC is roughly proportional to $Re^{0.8}$ inside the cold plate. Thus, the thermal resistance drops appreciably with the flowrate as shown in Fig. 16. Also shown in Fig. 16, the



(a) Temperature distribution of the cold plate placing in air ambience.



(b) Temperature distribution of the cold plate placing in fluid.

Fig. 13. Temperature distribution at T_{case} cross section (Side-open cold plate: Type B).

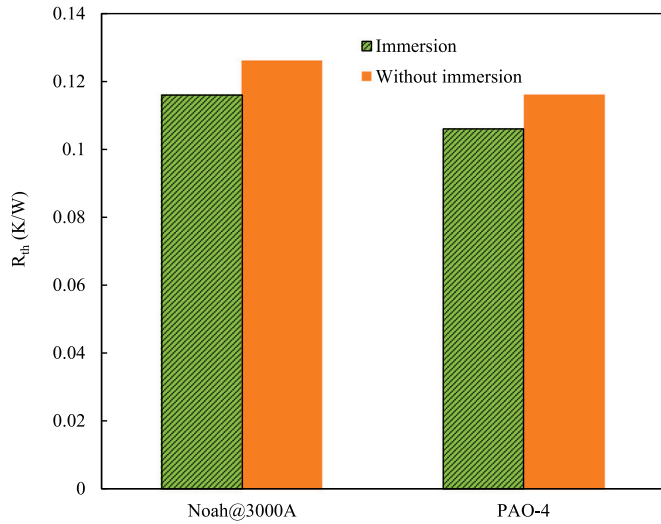


Fig. 14. The heat dissipation performance of the cold plate with or without immersion (Enclosed cold plate: Type B, Flowrate = 1 LPM).

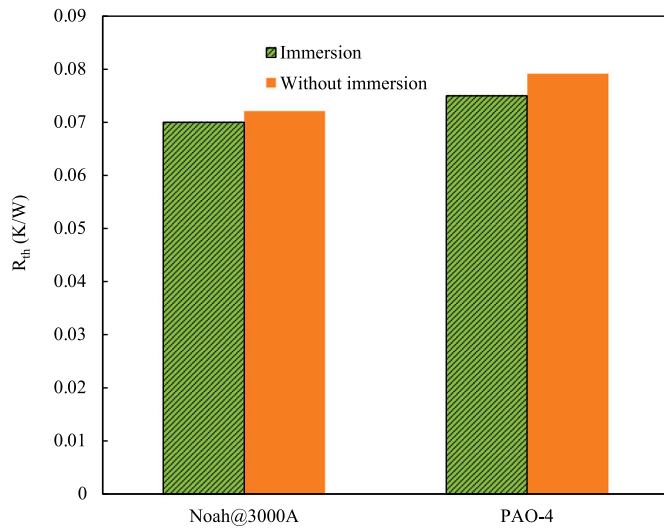


Fig. 15. The heat dissipation performance of the cold plate with or without immersion (Enclosed cold plate: Type B, Flowrate = 3 LPM).

Dual-open cold plate shows the least thermal resistance, followed by Enclosed cold plate, and the side-open one reveals the highest thermal resistance. This is because the shorter traveling distance for dual-open design provides a lower temperature difference between exits and heat source. The Enclosed cold plate shows 10% lower thermal resistance than that of side-open one because it provides sudden contraction at the outlet that promote the turbulence effect. Specifically, when Noah@3000 A was the used, the Dual-open cold plate displayed the lowest thermal resistance across all flow rates, averaging an 8.5% and 16.5% reduction in thermal resistance compared to the fully enclosed and side-open designs, respectively.

Conversely, the drop of thermal resistance against flowrate is less pronounced for PAO-4. This is because all the inlet Reynolds number falls well within the laminar flow regime. In this regard, the heat transfer coefficient is about proportional to $Re^{0.5}$, and a mild decline of thermal resistance vs. flowrate is shown in Fig. 17. It is also interesting to know that the Enclosed cold plate shows the least thermal resistance and the thermal resistance of side-open and dual-open is comparable. The smallest thermal resistance is seen for Enclosed cold plate because it contains sudden contraction and sharp turning at the outlet. This

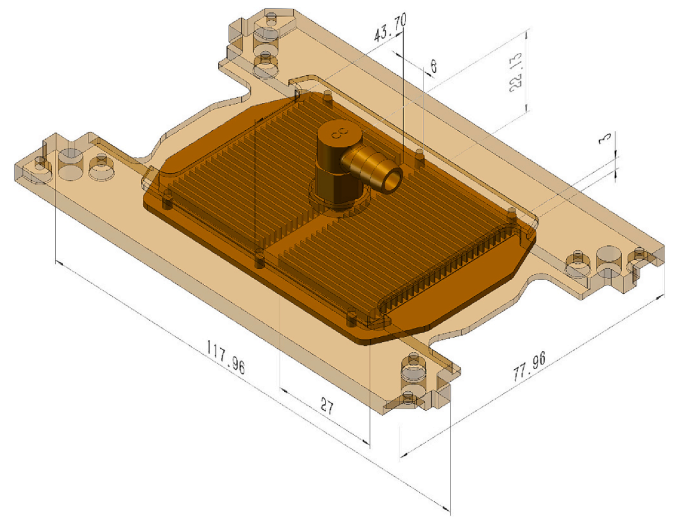


Fig. 16. The geometry of the impingement cold plate (Dual-open cold plate).

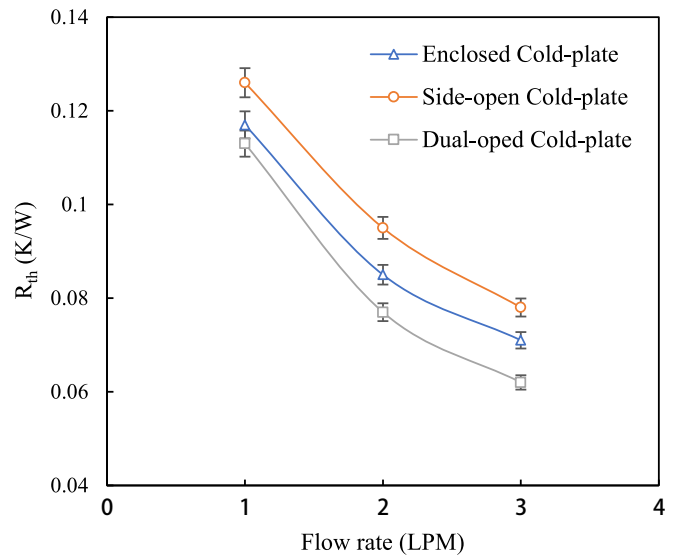


Fig. 17. Thermal resistance vs. flowrate for different Cold Plate Design (Type B, Noah@3000 A).

configuration yield rather turbulent flow at the outlet even though the main flow in the channel is still laminar. Consequently, the Enclosed cold plate shows the lowest thermal resistance. The side-open and Dual-open cold plate contains the same thermal resistance for all flowrates because they own the same inlet configuration and prevails at the same laminar flow regime alongside the channel, therefore no detectable difference in thermal resistance is observed. Specifically, when using PAO-4 as the working fluid, the fully Enclosed cold plate demonstrated superior performance, showing an average thermal resistance reduction of 10.1% and 10.5% compared to the dual-open and side-open designs, respectively.

When comparing the thermal performance between Enclosed cold plate and Side-open cold plate, the former consistently outperforms the latter, irrespective of the working fluid being utilized. This phenomenon is substantiated by the numerical simulations from Fig. 19 and Fig. 20. The simulation reveals that the flow field is accelerated especially near the center region that helps to reduce the T_{case} temperature (58.7 °C) at the heat source. The higher velocity near the center region is resultant from the contraction at the outlet. Note that the side-open one contains a T_{case} of 60.7 °C.

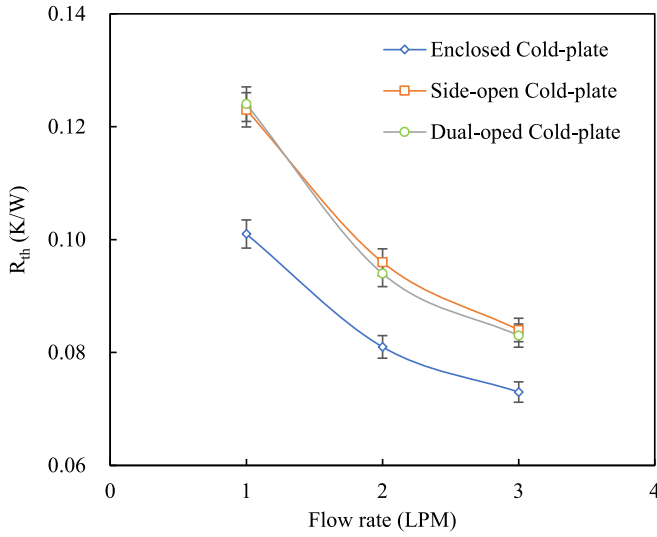


Fig. 18. Thermal resistance vs. flowrate for different cold plate design (Type B, PAO-4).

3.5. Effect of fin Design on the cold plate performance

To investigate the impact of fin structure on the thermal performance, experiments were conducted using three different fin dimensions having Enclosed cold plate whose geometric details are listed in Table 1. The results, illustrated in Fig. 21 and Fig. 22, indicate that regardless of whether Noah@3000 A or PAO-4 is used, a smaller fin pitch contribute to reduced thermal resistance. This is in contrast to employing traditional heat sink onto the chip. Using the traditional heat sink for typical immersion cooling system may cause severe fluid bypass which has been well elaborated by [28,40], and the bypass phenomenon worsens when the viscosity of the working fluid is raised (such as oils like PAO-6 & PAO-4). In dielectric fluids with lower viscosity, tightly spaced fins result in slightly enhanced thermal performance [40], while in fluids with higher viscosity, smaller fin pitch leads to unacceptable bypass and a rise of thermal resistance [28]. However, the present immersion cold plate eliminates the bypass effects entirely, thereby all thermal resistance reveals a pronounced drop of thermal resistance with smaller fin pitch, and this is applicable to both Naah@3000 A and PAO-

4. In additional to fully elimination of bypass effect, the small fin pitch also helps to reduce the effective hydraulic diameter which promote higher heat transfer coefficient accordingly.

4. Conclusions

This study investigates the use cold plate in single-phase immersion cooling system subject to dielectric fluid Noah@3000 A and PAO-4. Experimental tests are conducted in a 1 U sever having a TTV as the main heat source. The supplied flow rates range from 1 to 3 LPM. Numerical simulations are also performed for verification. The effects of flowrate, inlet and outlet configuration, and fin pitch of the cold plate are examined in details. The test cold plates include side-open, fully enclosed, and dual-open. The test results are also compared with traditional heat sink. Based on the foregoing discussions, the following conclusions are drawn:

- (1) In single-phase immersion cooling systems, the incorporation of cold plates substantially mitigates the fluid bypass effect, enhancing thermal performance. Whether utilizing Noah@3000 A or PAO-4 as the working fluid, the thermal resistance markedly diminishes as the rise of flowrate. However, the drop of thermal resistance subject to flowrate for Noah@3000 A is more pronounced due to its corresponding higher Reynolds number and turbulent flow nature. For the same cold plate, the thermal resistance of Noah@3000 A is inferior to PAO-4 at a lower flowrate of 1 LPM, this is because the thermal conductivity of PAO-4 is far higher than Noah@3000 A. However, when the flowrate is increased to 3 LPM, the trend is reversed due to significant turbulent flow condition of Noah@3000 A.
- (2) Upon immersion of the cold plate in liquid, external convective heat transfer improves and enhances the performance when compared to non-immersed conditions. At a low flow rate, the thermal resistances for both Noah@3000 A and PAO-4 are reduced by 7.2% and 9.4%, respectively.
- (3) Unlike the conventional heat sink, smaller fin pitch always leads to reduction of thermal resistance for all immersion cold plate. This is because the immersion cold plate eliminates the bypass entirely. The outlet also imposes considerable effect on the thermal resistance. For PAO-4, the Enclosed cold plate shows the least thermal resistance, and the thermal resistance for Side-open and Dual-open cold plate is very small. Conversely, in

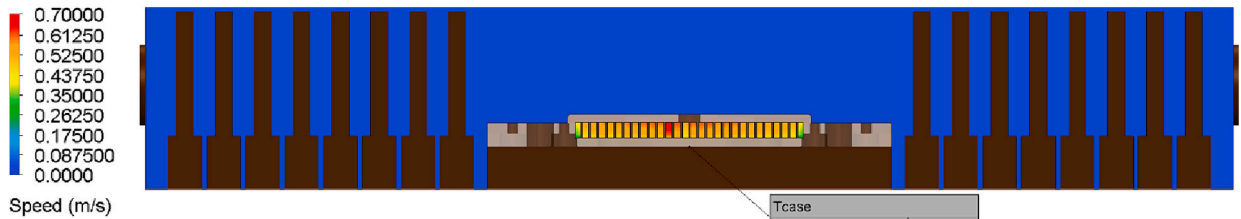


Fig. 19. Simulated flow field distribution of Enclosed cold plate. (Type B, Flowrate = 3 LPM).

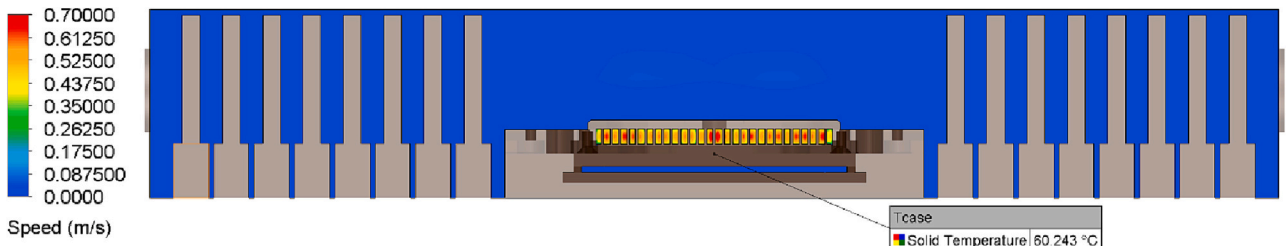


Fig. 20. Simulated flow field distribution of Side-open cold plate (Type B, Flow rate = 3 LPM).

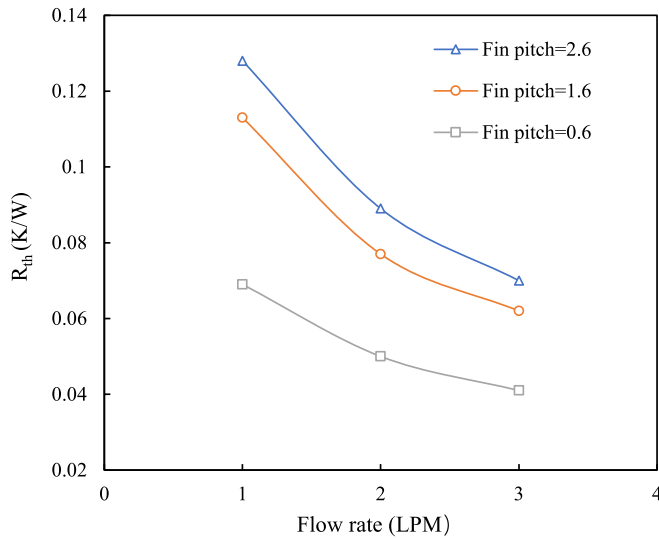


Fig. 21. Effect of fin pitch on the cold plate performance (Enclosed cold plate, Noah@3000 A).

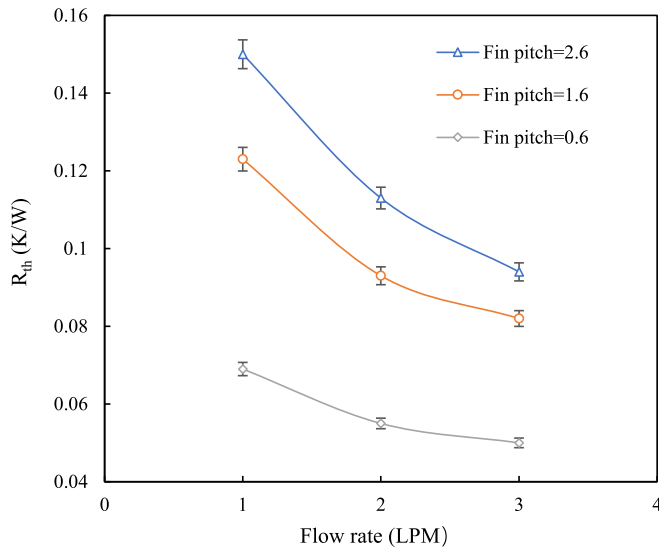


Fig. 22. Effect of fin pitch on the cold plate performance (Enclosed cold plate, PAO-4).

Noah@3000 A, Dual-open cold plates yield the least thermal resistance due to its smallest developing length. The difference in heat transfer behavior amid PAO-4 and Noah@3000 A is related to the laminar flow and turbulent flow characteristics, respectively.

Nomenclature

I	current (A)
V	voltage (V)
Q	input power (W)
R_{th}	thermal resistance ($^{\circ}\text{C W}^{-1}$)
T_{case}	temperature of the TTV ($^{\circ}\text{C}$)
T_{in}	inlet temperature of the fluid ($^{\circ}\text{C}$)
T_{out}	outlet temperature of the fluid ($^{\circ}\text{C}$)

Acronyms

ODP	ozone depletion potential
GWP	global warming potential
CFD	computational fluid dynamics
CPU	central processing unit
DC	data center
LPM	liters per minute
PAO	polyalphaolefin
PCB	printed circuit board
PUE	power usage effectiveness
RAM	random-access memory
TIM	thermal interface material
TTV	thermal test vehicle

CRediT authorship contribution statement

Yong-Dong Zhang: Writing – original draft, Data curation. **Yu-Chi Lin:** Validation, Investigation. **Chi-Chuan Wang:** Writing – review & editing, Project administration, Methodology.

Declaration of competing interest

Chi-Chuan Wang reports financial support was provided by Intel. Chi-Chuan Wang reports a relationship with Intel that includes: funding grants. Chi-Chuan Wang reports a relationship with National Science and Technology Council that includes: non-financial support. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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